GLOBAL LIFE SCIENCES LTD. | QUALITY CONTROL LABORATORY

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RECORD ID: DEV-2026-QC-1109 INCIDENT DATE: 09-Nov-2026

TESTING SUITE: HPLC Lab - Suite 3 STATUS: Under Investigation

PRODUCT AFFECTED: Paracetamol 500mg Tablets BATCH NUMBER: PCT-2026-084X

TEST PROCEDURE: Assay via HPLC-DAD COMPENDIAL REF: Ph. Eur. Monograph 2.2.29

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1.0 INCIDENT DESCRIPTION

During routine commercial release stability testing of Paracetamol 500mg Tablets (Batch: PCT-2026-084X), the High-Performance Liquid Chromatography (HPLC) autosampler sequence (Asset ID: HPLC-SYS-04, running Agilent OpenLab CDS) aborted automatically at Injection sequence #14 out of #25.

The instrument software triggered an automatic shutdown and logged a "System Pressure Error: Exceeded Upper Threshold Limit (>420 bar)". No analytical data was generated for the remaining sample set.

IMMEDIATE REACTION BY ANALYST (Smita Maharana):

- The chromatographic sequence was immediately paused within the software to preserve the system state.

- The sample tray was verified to remain intact; active vials were transferred to a monitored refrigerator (2-8°C) to prevent sample degradation during downtime.

- No electronic files, methods, or incomplete integration logs were deleted, overwritten, or modified, maintaining complete data traceability (ALCOA+ alignment).

- The incident was immediately escalated to the QC Laboratory Supervisor for a formal Phase Ia Laboratory Investigation.

2.0 PHASE IA LABORATORY ASSIGNMENT & ASSESSMENT CHECKLIST

[PASS] 1. DATA INTEGRITY CHECK: Were all electronic chromatographic logs, raw injections, and instrument error codes saved and left completely unaltered?

[PASS] 2. CALCULATION & GLASSWARE AUDIT: Were all volumetric dilution steps, mathematical multiplication factors, and sample weight transcriptions re-calculated and verified to be free of human mathematical error?

[PASS] 3. INSTRUMENT STATUS VERIFICATION: Was the physical column checked for visible leaks, fittings examined for crimps, and pump seals monitored for micro-voiding?

[FAIL] 4. MOBILE PHASE & FILTRATION RE-VERIFICATION: Was the mobile phase preparation log audited?

INVESTIGATION FINDING:

An audit of the reagent preparation log revealed that the Phosphate Buffer Mobile Phase (Batch: MP-1109) was filtered using a standard 0.45µm membrane filter instead of the specific 0.22µm nylon membrane filter mandated by the product's validated testing method parameters. Particulate matter from the buffer precipitated under high pressure, clogging the inlet frit of the analytical column, causing the pressure spike.

3.0 ROOT CAUSE ANALYSIS (RCA) - THE 5 WHYS

1. Why did the HPLC sequence automatically abort?

-- The instrument safety interlock cut power because system pressure exceeded 420 bar.

2. Why did the system pressure exceed the safety limit?

-- Particulate blockages built up rapidly at the inlet frit of the analytical column.

3. Why did particulate matter accumulate at the column inlet?

-- Micro-particles precipitated out of the mobile phase buffer during the high-pressure gradient run.

4. Why were micro-particles present in the mobile phase buffer?

-- The mobile phase was processed through an incorrect filter membrane pore-size (0.45µm instead of 0.22µm).

5. Why did the analyst use a 0.45µm filter instead of a 0.22µm membrane?

-- The analyst referenced a generic laboratory filtration SOP instead of cross-referencing the specific, validated product Testing Specification Sheet.

FINAL ROOT CAUSE CATEGORY: Human Error / Procedural Adherence Failure.

4.0 CORRECTIVE AND PREVENTIVE ACTION (CAPA) PLAN

CORRECTION (Immediate Fix):

- Safely purge the HPLC system line, replace the contaminated column guard filter, and flush the system with 10% Methanol.

- Prepare a fresh batch of mobile phase buffer, ensuring filtration through a certified 0.22µm nylon membrane.

- Re-run the system suitability sequence to confirm baseline stability.

CORRECTIVE ACTION (To Prevent Near-Term Recurrence):

- Issue a mandatory re-training record for the analytical chemist on SOP-QC-089 (Mobile Phase Preparation and Handling) with an emphasis on method-specific filtration thresholds.

PREVENTIVE ACTION (Long-Term Structural Safeguard):

- Update the Laboratory Information Management System (LIMS) software architecture. Program a mandatory electronic block: The HPLC sequence for Paracetamol Assay cannot be initiated until the analyst checks an electronic confirmation box confirming: "Verified 0.22µm filtration execution."

5.0 INVESTIGATION SIGN-OFFS

Prepared By: Smita Maharana, MSc (QC Analyst) Date: 09-Nov-2026

Reviewed By: _____ (QC Supervisor) Date: _____

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